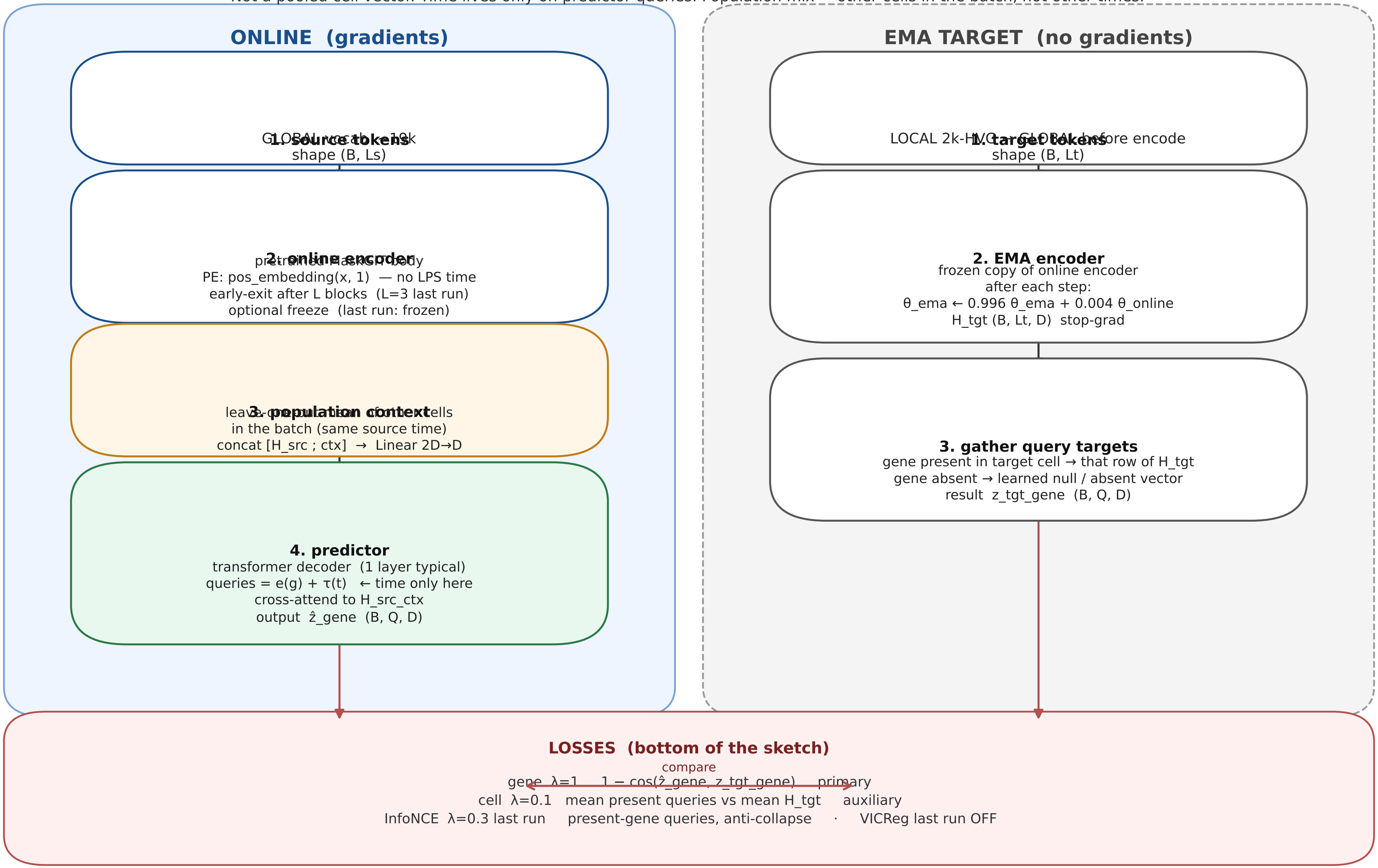
Gene-Query JEPA

Given a cell at source time, predict each queried gene's contextual embedding at a later LPS time.
Not a pooled cell vector. Time lives only on predictor queries. Population mix = other cells in the batch, not other times.



Sketch notes

Honesty metric (side note, not a training loss)

gene_gap_vs_copy_src = cos(prediction, target) − cos(source gene embedding, target)
Must be > 0. Copying the source also drops the loss. Do not trust a falling total_loss alone.

Query mix per cell (Q questions, typical Q = 64)

~50% genes present in BOTH source and target
~30% genes present ONLY in the target (induction; biologically most interesting)
rest   absent decoys; correct answer is the learned null vector

Shapes

B = cells in the mini-batch   Ls / Lt = padded source / target gene-sequence lengths
Q = gene queries per cell   D = 768 (pretrained MaskGIT encoder width)

Do not draw

PerturbGen's other-time decoder context (concatenating other timepoints into the decoder).
This JEPA only mixes other cells at the source time.

Typical last full-run knobs (labels only)

freeze encoder = true   L = 3   predictor layers = 1   Q = 64 mixed
λ gene / cell / contr = 1.0 / 0.1 / 0.3   VICReg off
split = False (all LPS cells)   20 epochs   D = 768

Code map

perturbgen/Modules/gene_query_jepa.py	model
perturbgen/Model/gene_query_jepa_trainer.py	sampler + losses
perturbgen/Modules/jepa_scmaskgit.py	pretrained encoder wrapper
docs/examples/train_gene_query_jepa.py	toy | full train